Reinforcement Learning Pt 1/2 (Value estimation)

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13.1 The Reinforcement Learning (RL) setting

Reminder:

- Data in supervised learning: $\left\{\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N, \underline{\mathbf{y}}_T^{(p)} \in \mathbb{R}^M\right\}_{\alpha=1}^{p}$
- Data in **un**supervised learning: $\left\{\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N\right\}_{\alpha=1}^{p}$

Data in Reinforcement Learning:

A sequence of random states $\underline{\mathbf{x}}$

resulting from taking actions $\underline{\mathbf{a}}$ over time

which result in some reward r.

- a state $\underline{\mathbf{x}} \in \mathcal{X}$ or $\underline{\mathbf{x}} \in \mathbb{R}^N$, e.g. $\mathcal{X} := \{\underline{\mathbf{x}}_1, \ldots, \underline{\mathbf{x}}_S\} \subset \{0,1\}^S$ (1-out-S encoding)
- an action $\underline{\mathbf{a}}$ which can be taken by the agent:

  $\underline{\mathbf{a}} \in \mathcal{A}$ or $\underline{\mathbf{a}} \in \mathbb{R}^M$, e.g. $\mathcal{A} := \{\underline{\mathbf{a}}_1, \ldots, \underline{\mathbf{a}}_A\} \subset \{0,1\}^A$ (1-out-A encoding)
- a reward $r \in \mathbb{R}$ or $r \in \{0,1\}$, e.g. $r \in \{$"cheese", "no cheese"$\}$.

Each time step describes what reward was received when performing some action while in some state.

The sequence we observe becomes:

$$\left\{\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}, r^{(t)}\right\}_{t=0}^{p} = \left(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)}, r^{(0)}\right), \ldots, \left(\underline{\mathbf{x}}^{(p)}, \underline{\mathbf{a}}^{(p)}, r^{(p)}\right) \quad (13.1.1)$$

13.1.1 The Transition Model

Markov property

The probability of transitioning from state $\underline{\mathbf{x}}_i$ to $\underline{\mathbf{x}}_j$ only depends on state $\underline{\mathbf{x}}_i$ and the action $\underline{\mathbf{a}}_k$ that was taken. It does not depend on earlier history.

This leads to the following transition model (transition "matrix" or "volume"):

$$P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \quad (13.1.2)$$

which

- measures the probability to end up in $\underline{\mathbf{x}}_j$ after choosing $\underline{\mathbf{a}}_k$ when in $\underline{\mathbf{x}}_i$,
- is a stationary distribution

$$P(\underline{\mathbf{x}}_j^{(t+1+\tau)} | \underline{\mathbf{x}}_i^{(t+\tau)}, \underline{\mathbf{a}}_k^{(t+\tau)}) = P(\underline{\mathbf{x}}_j^{(t+1)} | \underline{\mathbf{x}}_i^{(t)}, \underline{\mathbf{a}}_k^{(t)}), \quad \tau \in \{0, 1, \ldots\} \quad (13.1.3)$$

and where

$$P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \geq 0 \quad \forall i, j, k \quad (13.1.4)$$

$$\text{and} \sum_j P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 1 \quad \forall i, k \quad \text{i.e. every "row" sums to 1} \quad (13.1.5)$$

Q 13.1.1: *What would* $\sum_{\substack{j \\ j \neq i}} P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 0 \ \forall\, k$ *imply?*

- $\underline{\mathbf{x}}_i$ is a state which leads nowhere else, regardless of the action. It is a *terminal state*.

Q 13.1.2: *What does an entry* $P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 0$ *represent?*

- When at $\underline{\mathbf{x}}_i$, taking action $\underline{\mathbf{a}}_k$ can never lead to $\underline{\mathbf{x}}_j$.

Q 13.1.3: *What does an entry* $P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 0 \ \forall\, k$ *represent?*

- It is impossible to go from $\underline{\mathbf{x}}_i$ to $\underline{\mathbf{x}}_j$, regardless of the action.

Q 13.1.4: *What does an entry* $P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 1$ *represent?*

- When at $\underline{\mathbf{x}}_i$, taking action $\underline{\mathbf{a}}_k$ always leads to $\underline{\mathbf{x}}_j$.

Markov chain

a sequence of random states $\underline{\mathbf{x}}$ with a *Markov property*.

13.1.2 The Policy

Policy

Representation of the agent[1]. It measures the distribution over actions given states.

- map states to actions, i.e. the probability of doing $\underline{\mathbf{a}}_k$ when in $\underline{\mathbf{x}}_i$:

$$\text{policy } \pi := \pi(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i) \tag{13.1.6}$$

- the mapping can be stochastic or deterministic
- the policy is stationary. That the policy changes during the training process does not contradict this. It is regarded as stationary because once trained and deployed, the way the agent decides does not change over time.

[1] Without a policy we would only have a Markov Reward Process. Having a policy (wanting to find a policy) turns the Markov Reward Process into a Markov Decision Process.

13.2 Markov Decision Process (MDP)

A formalized description of the environment in RL[2]

Assumption of MDP

The environment is fully observable. There is a complete characterization of what's going on and where everything is.

The other case would be *partially observable* MDP (POMDP)

What do we want to do?

Maximize, not just immediate rewards, but the *cumulative sum of rewards* or "return".

$$return(t) := \sum_{k=0}^{\infty} \gamma^k r(\underline{\mathbf{x}}^{(t+k+1)}, \underline{\mathbf{a}}^{(t+k+1)}) = \sum_{k=0}^{\infty} \gamma^k r^{(t+k+1)}, \qquad (13.2.1)$$

where $\gamma \in [0, 1)$ is referred to as the *discount factor*.

About the *discount factor* γ:

- It is possible for γ to be $= 1$.

  Q 13.2.1: *Under what condition can γ be $= 1$*

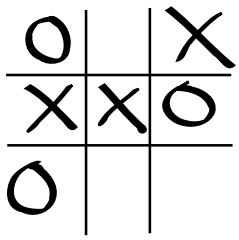


Figure 13.2.1: Some games can always be completeted in a finite number of steps.

  - γ can be $= 1$ if the length of the sequence is guaranteed to be finite.

- Q 13.2.2: *What does $\gamma \to 0$ imply?*

  - preference for short term rewards

- Q 13.2.3: *What does $\gamma \to 1$ imply?*

  - emphasis on long-term returns, indifference to delay.

Fig. 13.2.2 illustrates how the choice of γ controls how fast the discount factors decay over time.

[2]Recommended reading (Sutton et al., 1998) also video lectures by David Silver

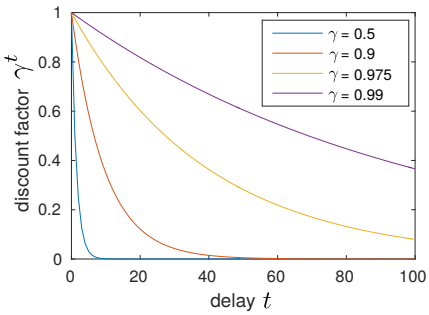


Figure 13.2.2: modulating returns with different discount factors

Example: big reward further away vs. closer smaller reward

Fig. 13.2.3) illustrates a simple environment. The mouse can move one tile at a time to the **R**ight or to the **L**eft. We'll compare the return of two different strategies for different discount factors.

Figure 13.2.3: Environment with two rewards placed at different distances from the initial position of the mouse.

strategy	γ	$return(0)$	
R,R,...	$\frac{1}{2}$	$1 \cdot 0 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 1 + \ldots$	≥ 0.25
L,L,L,...		$1 \cdot 0 + \frac{1}{2} \cdot 0 + \frac{1}{4} \cdot 0 + \frac{1}{8} \cdot 3 + \ldots$	$\geq \mathbf{0.375}$
R,R,...	0.7	$1 \cdot 0 + 0.7 \cdot 0 + 0.7^2 \cdot 1 + \ldots$	≥ 0.49
L,L,L,...		$1 \cdot 0 + 0.7 \cdot 0 + 0.7^2 \cdot 0 + 0.7^3 \cdot 3 + \ldots$	$\geq \mathbf{1.029}$
R,R,...	0.2	$1 \cdot 0 + 0.2 \cdot 0 + 0.2^2 \cdot 1 + \ldots$	$\geq \mathbf{0.04}$
L,L,L,...		$1 \cdot 0 + 0.2 \cdot 0 + 0.2^2 \cdot 0 + 0.2^3 \cdot 3 + \ldots$	≥ 0.024

Table 1: Comparing returns under different discount factors.

Q 13.2.4: *Why do we discount future rewards? What criteria go into choosing γ?*

- Uncertainty of the future
- Imperfections in the model (we don't trust its decisions completely)
- mathematically convenient, the sum does not explode
- avoids ∞ returns due to possible cycles in our MDP
- behavioral arguments (animals and humans seem to apply a similar discounting) "A bird in the hand is worth two in the bush"
- financially realistic (accounts for inflation)

13.2.1 The value function

The (state) value function $\widehat{=}$ the expected sum of discounted future rewards

Q 13.2.5: *What does the value function represent?*

- The long-term value of some state $\underline{\mathbf{x}}$,
- the expected return from starting in $\underline{\mathbf{x}}^{(0)}$,
- we have a preference for high expectations and MDPs maximize this quantity

For a fixed policy π:

- a **value function** measures the quality of a policy π in state $\underline{\mathbf{x}}^{(0)}$
  - $V^{\pi}(\underline{\mathbf{x}}^{(0)})$ is the *expected infinite sum of discounted future rewards*

$$V^{\pi}(\underline{\mathbf{x}}^{(0)}) \;=\; \mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^{t}\, r(\underline{\mathbf{x}}^{(t)},\underline{\mathbf{a}}^{(t)}) \;\middle|\; \begin{array}{c}\underline{\mathbf{a}}^{(t)} \sim \pi(\cdot\,|\,\underline{\mathbf{x}}^{(t)})\\ \underline{\mathbf{x}}^{(t+1)} \sim P(\cdot\,|\,\underline{\mathbf{x}}^{(t)},\underline{\mathbf{a}}^{(t)})\end{array}\right], \quad \gamma \in [0,1)\,. \tag{13.2.2}$$

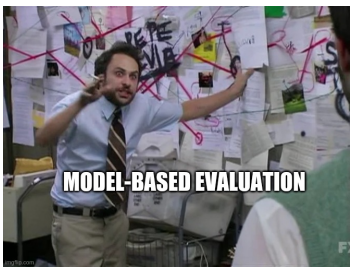
In other textbooks you may find:

$$V^{\pi}(\underline{\mathbf{x}}_i) = \overbrace{V_i^{\pi}}^{\text{shorthand}} = \mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^{t} r(\underline{\mathbf{x}}^{(t)},\underline{\mathbf{a}}^{(t)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right],\; i = 1,\ldots,S \tag{13.2.3}$$

$\mathbb{E}[\cdot]$ is w.r.t $\{\underline{\mathbf{x}}^{(t)},\underline{\mathbf{a}}^{(t)}\}$

13.3 Model-based evaluation of the policy

Model-based $\widehat{=}$ offline-planning. We are told everything about the environment,so we can simulate "looking ahead" and determine if one policy is better than another.



13.3.1 The Bellman equation

The Bellman equation relates the value of a state with the values of its successor states, given some policy π.

The policy is fixed beforehand, i.e. V "follows" the policy π.

$$V_i^\pi = \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t \overbrace{r(\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)})}^{\substack{\text{shorthand} \\ =:r^{(t)}}} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big], \; i = 1, \dots, S \tag{13.3.1}$$

$$= \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r^{(t)} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \tag{13.3.2}$$

$$= \mathbb{E}\Big[\gamma^0 \underbrace{r^{(0)}}_{\substack{\text{immediate} \\ \text{reward}}} + \gamma^1 r^{(1)} + \gamma^2 r^{(2)} + \dots \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \tag{13.3.3}$$

$$= \mathbb{E}\Big[r^{(0)} + \underbrace{\gamma\big(r^{(1)} + \gamma r^{(2)} + \dots\big)}_{\substack{\text{discounted value of} \\ \text{successor state}}} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \tag{13.3.4}$$

$$= \mathbb{E}\Big[r^{(0)} + \gamma V^\pi(\underline{\mathbf{x}}^{(1)}) \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \tag{13.3.5}$$

split "expectation of sum" into "sum of expectations":

$$= \mathbb{E}\Big[r^{(0)} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] + \mathbb{E}\Big[\gamma V^\pi(\underline{\mathbf{x}}^{(1)}) \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \tag{13.3.6}$$

$$= \mathbb{E}\Big[r^{(0)} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] + \gamma\, \mathbb{E}\Big[V^\pi(\underline{\mathbf{x}}^{(1)}) \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \tag{13.3.7}$$

Q 13.3.1: *What does the $\mathbb{E}[\cdot]$ boil down to?*

$$V_i^\pi = \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\substack{\text{“controlled” reward function} \\ \text{(immediate reward)}=:r_i^\pi}} + \underbrace{\gamma \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) \sum_{j=1}^{S} P(\underline{\mathbf{x}}_j \,|\, \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \overbrace{V_j^\pi}^{=V^\pi(\underline{\mathbf{x}}_j)}}_{\text{expected discounted value of successor state}} \tag{13.3.8}$$

$$= \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\text{“controlled” reward function } r_i^\pi} + \gamma \sum_{j=1}^{S} \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) P(\underline{\mathbf{x}}_j \,|\, \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\text{“controlled” transition model } P_{ij}^\pi} V_j^\pi \tag{13.3.9}$$

Thus, the Bellman equation can be expressed using:

$$V_i^\pi = \underline{\mathbf{r}}_i^\pi + \gamma \underline{\mathbf{P}}_{ij}^\pi V_j^\pi \tag{13.3.10}$$

We can also use vector notation to represent the value of all states collectively:

$$\underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi \;=:\; \overbrace{\hat{B}^\pi[\underline{\mathbf{v}}^\pi]}^{\text{the Bellman operator}}, \tag{13.3.11}$$

where $\underline{\mathbf{v}}^\pi \in \mathbb{R}^S$ is a vector containing all values $V^\pi(\underline{\mathbf{x}}_i)\ \forall i$,

$$\text{with} \underbrace{\begin{cases} r_i^\pi := \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i)\, r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \\ P_{ij}^\pi := \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i)\, P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \end{cases}}_{\text{“controlled” models } \underline{\mathbf{r}}^\pi \in \mathbb{R}^S \text{ and } \underline{\mathbf{P}}^\pi \in \mathbb{R}^{S \times S}} \tag{13.3.12}$$

13.3.2 Finding the value

There are two approaches to finding $\underline{\mathbf{v}}^\pi$:

13.3.2.1 - analytically

13.3.2.2 - through fixed-point iteration

13.3.2.1 Analytical solution of the Bellman equation

Essentially solve for $\underline{\mathbf{v}}^\pi$ (get <u>all</u> $\underline{\mathbf{v}}^\pi$ on one side):

$$\underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi \tag{13.3.13}$$

$$\underline{\mathbf{I}}\, \underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi \tag{13.3.14}$$

$$\underline{\mathbf{I}}\, \underline{\mathbf{v}}^\pi - \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi \tag{13.3.15}$$

$$\left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^\pi\right) \underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi \tag{13.3.16}$$

$$\underline{\mathbf{v}}^\pi = \left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^\pi\right)^{-1} \underline{\mathbf{r}}^\pi \tag{13.3.17}$$

Q 13.3.2: *When is the term* $\left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^\pi\right)$ *invertible?*

1. $|\lambda_k| \leq 1$ for all eigenvalues λ_k of transition matrix $\underline{\mathbf{P}}^\pi$
2. $\gamma < 1$

$\Rightarrow$ it is always invertible

Q 13.3.3: *Any disadvantages to the analytical solution?*

- It could be very costly for very large state spaces.

13.3.2.2 Value iteration

Value iteration: a fixed-point iteration for finding $\underline{\mathbf{v}}^\pi$

- A fixed point: $x = g(x) = g(g(\ldots g(x)\ldots))$

  Example:

$$g(x) = x^2 - 3x + 4 \tag{13.3.18}$$
$$g(2) = 2^2 - 3 \cdot 2 + 4 = 4 - 6 + 4 = 2 \tag{13.3.19}$$

- Fixed point iteration:

$$x^{(t+1)} = g(x^{(t)}), t = 0, 1, 2, \ldots \tag{13.3.20}$$

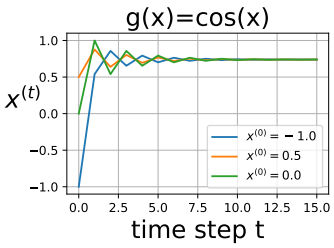


Figure 13.3.1: Fixed point iteration example

$$\underline{\tilde{\mathbf{v}}}^{\pi(t+1)} = \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\tilde{\mathbf{v}}}^{\pi(t)} \tag{13.3.21}$$

- The "˜" is to denote that it has not necessarily converged at time step t
- initialize $\underline{\tilde{\mathbf{v}}}^{\pi(0)}$ with some state $\underline{\tilde{\mathbf{v}}}^{\pi(0)} \in \mathbb{R}^S$
- value iteration converges in the limit $t \to \infty$
- The speed of convergence depends on γ

Q 13.3.4: *How does convergence depend on γ?*

- small γ: fast convergence
- $\gamma \to 1$: slow convergence, slower than the analytical solution
- We start *value iteration* assuming $\underline{\tilde{\mathbf{v}}}^{\pi(0)}$ is the optimal value function
- Use the one-step *look-ahead* (i.e. next step) to work our way *backwards*

$$\hat{B}^\pi[\underline{\mathbf{v}}^\pi] := \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underbrace{\underline{\mathbf{v}}^\pi}_{\text{look-ahead}}$$

Analytical solution vs. value iteration:

- complexity:
  - Analytical solution: $O(S^3)$
  - value iteration: $O(S^2)$
- intermediate solutions of *value iteration* don't necessarily belong to valid policies. We have to wait for convergence.

13.3.3 Finding the optimal policy

13.3.3.1 Comparing two policies

A policy π is regarded as better or equal to another policy π' if π yields an expected return which is greater or equal to that of the other policy π' for all possible states. That is,

$$\pi \overset{\text{“better” or equally “good”}}{\geq} \pi' \quad \textbf{iff} \quad V^{\pi}(\underline{\mathbf{x}}) \geq V^{\pi'}(\underline{\mathbf{x}}), \quad \forall\, \underline{\mathbf{x}} \in \mathcal{X} \tag{13.3.22}$$

Multiple *optimal policies* can exist. A policy is optimal if it maximizes the state value function[3]:

$$V^{*}(\underline{\mathbf{x}}) = \max_{\pi} V^{\pi}(\underline{\mathbf{x}}), \quad \forall\, \underline{\mathbf{x}} \in \mathcal{X} \tag{13.3.23}$$

Optimal Policy

For all optimal policies π^*:

$$V^{\pi^*}(\underline{\mathbf{x}}) = V^{*}(\underline{\mathbf{x}}) \tag{13.3.24}$$

A policy π that maximizes the value function and yields $\underline{\mathbf{v}}^*$ is an optimal policy π^*.

13.3.4 Extracting the policy from an optimal value function

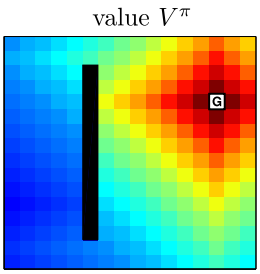


Figure 13.3.2: Extract optimal policy from optimal value function

$$\pi^{*}(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i) = \underset{\underline{\mathbf{a}}}{\operatorname{argmax}} \left(r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}) + \gamma \sum_{j=1}^{S} P(\underline{\mathbf{x}}_j|\underline{\mathbf{x}}_i, \underline{\mathbf{a}}) V^{*}(\underline{\mathbf{x}}_j) \right) \tag{13.3.25}$$

Q 13.3.5: *What is going on here?*

1. When at $\underline{\mathbf{x}}_i$, look at the actions $\underline{\mathbf{a}}$ that lead to all possible future states $\underline{\mathbf{x}}_j$.
2. Select action $\underline{\mathbf{a}}_k$ that maximizes the immediate reward + discounted successive value.

[3] For more information, see Ch 3.6 in (Sutton et al., 1998)

Q 13.3.6: *What are the implications of selecting the optimal policy this way?*

- The above does not scale well for large state spaces.
- The resulting π^* is a *greedy* policy, because it greedily selects using only the value function at hand.

13.4 References

References

Richard S Sutton, Andrew G Barto, et al. *Introduction to reinforcement learning*, volume 135. MIT press Cambridge, 1998.